

Coefficient spaces and an explicit Riemann map for the Stolz-domain question in arXiv:2303.03022

Autonomous mathematical investigation, lane 5

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Abstract

The open-question subsection of Fackler and Hauer asks for the spaces generated by products of their standard Ritt square functions, for a general fixed-coefficient representation, and for an explicit conformal map from their Stolz domain to the disc. We identify every product space exactly as a weighted analytic coefficient space and identify their algebraic sum. This also gives a sharp obstruction: the proposed products cannot represent every bounded analytic function, because all their finite sums extend to the whole unit disc. We then give and prove an explicit normalized Riemann map in terms of a generalized Chebyshev function. Finally, we show why simply pulling standard monomial square functions back through this map cannot solve the remaining square-function problem. Thus the coefficient-space and conformal map parts are complete; the request for a genuinely new elementary square family remains open.

1 Question and answer

For $\omega > 1$, set

$$\text{Stolz}_\omega = \left\{ z \in \mathbb{D} : \frac{|1-z|}{1-|z|} < \omega \right\}.$$

The source defines, for integers $m \geq 1$ and $k \geq 1$,

$$\varphi_m(k, z) = k^{m-\frac{1}{2}}(1-z)^m z^{k-1}$$

and asks to identify

$$\mathcal{S}_{m_1, m_2} = \left\{ \sum_{k \geq 1} a_k \varphi_{m_1}(k, z) \varphi_{m_2}(k, z) : (a_k) \in \ell_\infty \right\},$$

their sum in $H^\infty(\text{Stolz}_\omega)$, and the conformal map of Stolz_ω onto $\mathbb{D}$.

For an integer $s \geq 0$, introduce the weighted coefficient space

$$\mathcal{A}_s = \left\{ A(w) = \sum_{n \geq 0} c_n w^n : \sup_{n \geq 0} \frac{|c_n|}{(n+1)^s} < \infty \right\}.$$

Also write $D = w \frac{d}{dw}$ and

$$H_{\text{coeff}}^\infty = \left\{ \sum_{n \geq 0} b_n w^n : (b_n) \in \ell_\infty \right\}.$$

Theorem 1 (Complete coefficient and conformal-map subresults). *Let $m_1, m_2 \geq 1$ and $M = m_1 + m_2$.*

1. *The space depends only on M , and exactly*

$$\begin{aligned}\mathcal{S}_{m_1, m_2} &= \mathcal{S}_M = \{(1-z)^M A(z^2) : A \in \mathcal{A}_{M-1}\} \\ &= (1-z)^M (D+1)^{M-1} H_{\text{coeff}}^\infty(z^2).\end{aligned}$$

Consequently the algebraic sum requested in the source is

$$\sum_{m_1, m_2 \geq 1}^{\text{alg}} \mathcal{S}_{m_1, m_2} = \left\{ \sum_{M \in F} (1-z)^M A_M(z^2) : F \subset \{2, 3, \dots\} \text{ finite, } A_M \in \mathcal{A}_{M-1} \right\}.$$

2. *This algebraic sum is a proper subspace of $H^\infty(\text{Stolz}_\omega)$. In particular, the fixed-coefficient formula proposed as (6.5) in the source cannot represent all of $H^\infty(\text{Stolz}_\omega)$, for any choices of m_1, m_2 .*

3. *Put*

$$a = \frac{1}{\omega}, \quad \beta = \arccos a, \quad \kappa = \frac{\pi}{2\beta}, \quad C(z) = \frac{1+z}{1-z}.$$

Let the generalized Chebyshev function be the branch

$$T_\kappa(w) = {}_2F_1\left(-\kappa, \kappa; \frac{1}{2}; \frac{1-w}{2}\right),$$

analytic on $\mathbb{C} \setminus (-\infty, -1]$. Equivalently, $T_\kappa(\cosh u) = \cosh(\kappa u)$ whenever $|\text{Im } u| < \beta$. Then

$$\boxed{F_\omega(z) = \frac{T_\kappa(C(z)) - 1}{T_\kappa(C(z)) + 1}} \tag{1}$$

is a conformal bijection from Stolz_ω onto $\mathbb{D}$. It is normalized by

$$F_\omega(0) = 0, \quad F'_\omega(0) = \kappa^2 > 0, \quad F_\omega\left(\frac{1-\omega}{1+\omega}\right) = -1, \quad \lim_{z \rightarrow 1} F_\omega(z) = 1.$$

2 Idea of the proof

The coefficient question collapses immediately after multiplying the two standard functions: the result is an even power series, multiplied by $(1-z)^M$, whose n th coefficient has exactly the allowed growth $(n+1)^{M-1}$. This not only identifies the spaces but exposes their common analytic-continuation property. A single pole placed in the part of the unit disc outside the closed Stolz domain then rules out the hoped-for universal fixed-coefficient representation.

For the conformal map, the Cayley transform turns the defining metric ratio into a difference of distances from the foci ± 1 . Hence the boundary becomes a hyperbola, parametrized by $\cosh(t \pm i\beta)$. The hyperbola domain is the quotient of the strip $|\text{Im } u| < \beta$ by $u \sim -u$. Scaling this strip to $|\text{Im } v| < \pi/2$ and applying $\cosh$ again maps the quotient to the right half-plane. This is precisely a generalized Chebyshev function; a final Cayley map gives the disc formula.

3 The coefficient spaces

The product defining the source space is

$$\varphi_{m_1}(k, z)\varphi_{m_2}(k, z) = k^{M-1}(1-z)^M z^{2k-2}. \quad (2)$$

Thus, if $n = k - 1$, the coefficient of w^n in the corresponding $A(w)$ is $(n+1)^{M-1}a_{n+1}$. This proves both set equalities in part 1, because

$$(D+1)^{M-1} \sum_{n \geq 0} b_n w^n = \sum_{n \geq 0} (n+1)^{M-1} b_n w^n.$$

Conversely, every $A \in \mathcal{A}_{M-1}$ produces a bounded function on Stolz_ω . Indeed, for $r = |z|$,

$$\begin{aligned} |(1-z)^M A(z^2)| &\leq C \|A\|_{\mathcal{A}_{M-1}} |1-z|^M \sum_{n \geq 0} (n+1)^{M-1} r^{2n} \\ &\leq C_M \|A\|_{\mathcal{A}_{M-1}} \left(\frac{|1-z|}{1-r^2} \right)^M \leq C_M \omega^M \|A\|_{\mathcal{A}_{M-1}}. \end{aligned}$$

Here $|1-z| < \omega(1-r)$ and $1-r^2 \geq 1-r$. Taking finite sums proves the asserted formula for the algebraic sum.

There is also a decisive obstruction to a universal representation. Every $A \in \mathcal{A}_s$ has radius of convergence at least one, hence every member of the algebraic sum extends holomorphically to all of $\mathbb{D}$. On the other hand, set

$$z_0 = -\frac{\omega}{\omega+1}.$$

This point lies in $\mathbb{D} \setminus \overline{\text{Stolz}_\omega}$, so $f(z) = (z - z_0)^{-1}$ belongs to $H^\infty(\text{Stolz}_\omega)$. If f agreed on Stolz_ω with a function G holomorphic on $\mathbb{D}$, the identity theorem on the connected punctured disc $\mathbb{D} \setminus \{z_0\}$ would give $G(z) = (z - z_0)^{-1}$ there. Continuity at z_0 would then force $0 = (z - z_0)G(z)|_{z=z_0} = 1$, a contradiction. This proves part 2.

Notice that the conclusion concerns the algebraic sum asked for in the source; no assertion about its norm closure is needed.

4 Proof of the explicit conformal map

The Cayley map C sends $\mathbb{D}$ to the right half-plane. If $w = C(z)$, then

$$z = \frac{w-1}{w+1}, \quad \frac{|1-z|}{1-|z|} = \frac{2}{|w+1| - |w-1|}.$$

Therefore $C(\text{Stolz}_\omega)$ is the component to the right of the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{1-a^2} = 1, \quad a = \frac{1}{\omega}.$$

Writing $\beta = \arccos a$, its two boundary arcs have the parametrization

$$w = \cosh(t \pm i\beta) = a \cosh t \pm i\sqrt{1-a^2} \sinh t, \quad t \in \mathbb{R}.$$

Let $S_\beta = \{u : |\text{Im } u| < \beta\}$. The map $u \mapsto \cosh u$ maps S_β onto $C(\text{Stolz}_\omega)$; its only identification inside this strip is $u \sim -u$. Similarly, $v \mapsto \cosh v$ maps $S_{\pi/2}$ onto the right half-plane with the

same identification. Multiplication by $\kappa = \pi/(2\beta)$ is a conformal bijection $S_\beta \rightarrow S_{\pi/2}$ and respects $u \sim -u$. It therefore descends to a conformal bijection

$$w = \cosh u \mapsto \cosh(\kappa u) = T_\kappa(w)$$

from $C(\text{Stolz}_w)$ onto the right half-plane.

This quotient argument also resolves the apparent branch at $w = 1$: locally $w - 1 = u^2/2 + O(u^4)$ and $T_\kappa(w) - 1 = \kappa^2 u^2/2 + O(u^4)$, so $T'_\kappa(1) = \kappa^2$. The hypergeometric expression in the theorem is the single-valued analytic continuation across $(-1, 1)$ and has its first branch obstruction only at $(-\infty, -1]$, well outside the hyperbola domain. Finally, $(q - 1)/(q + 1)$ maps the right half-plane onto $\mathbb{D}$. Composing these three bijections gives (1) and the stated normalizations.

An explicit inverse, with the strip branch of arccosh , is

$$F_\omega^{-1}(\zeta) = \frac{\cosh\left(\kappa^{-1} \text{arccosh} \frac{1+\zeta}{1-\zeta}\right) - 1}{\cosh\left(\kappa^{-1} \text{arccosh} \frac{1+\zeta}{1-\zeta}\right) + 1}.$$

Changing the permitted inverse branch only changes the argument by a sign, which disappears under $\cosh$.

5 Why the map does not by itself solve the square-function problem

One tempting upgrade is to replace z in the standard family by $\zeta = F_\omega(z)$. It fails at the level of the required pointwise ℓ_2 bound. For $m \geq 1$,

$$\sum_{k \geq 1} k^{2m-1} |1 - \zeta|^{2m} |\zeta|^{2k-2} \asymp_m \left(\frac{|1 - \zeta|}{1 - |\zeta|^2} \right)^{2m}.$$

As ζ approaches any point of $\partial\mathbb{D} \setminus \{1\}$, the right side diverges. The conformal map sends the whole curved Stolz boundary to $\partial\mathbb{D}$, so the cancellation that made the original Ritt family bounded only at its distinguished vertex is lost.

There is a more general diagonal obstruction. Suppose two disc families have the form

$$\psi_j(\zeta) = u_j \zeta^{p_j}, \quad \tilde{\psi}_j(\zeta) = v_j \zeta^{q_j}, \quad p_j, q_j \geq 0,$$

and both are uniformly pointwise square summable. Letting $r \uparrow 1$ and using monotone convergence gives $u, v \in \ell_2$, hence $(u_j v_j) \in \ell_1$. Every bounded-coefficient sum $\sum_j a_j \psi_j \tilde{\psi}_j$ therefore has absolutely summable Taylor coefficients, even when exponents repeat. It lies in the Wiener algebra and extends continuously to $\mathbb{D}$. This cannot cover $H^\infty(\mathbb{D})$: for example

$$S(\zeta) = \exp\left(-\frac{1 + \zeta}{1 - \zeta}\right)$$

is bounded and analytic on $\mathbb{D}$ but has no continuous extension at 1. Thus a successful explicit family must be genuinely non-diagonal and localized simultaneously in scale and boundary position.

There is a constructive, but not elementary, fallback. The later paper arXiv:2410.22006 constructs contour atoms Φ_j with $\sup_z \sum_j |\Phi_j(z)|^{1/2} < \infty$ and bounded-coefficient Franks–McIntosh representations. Formula (1) makes their canonical factors specifiable: factor $\Phi_j \circ F_\omega^{-1} = I_j O_j$ in the disc and set

$$\psi_j = (I_j O_j^{1/2}) \circ F_\omega, \quad \tilde{\psi}_j = O_j^{1/2} \circ F_\omega.$$

The outer factor has the standard boundary-integral formula, and the maximum principle transfers the $\sum |\Phi_j|^{1/2}$ bound to both factor families. This supplies explicit coordinates for the known contour construction, but does not produce the simple Ritt-adapted square family sought in the source.

6 Verification and literature boundary

The accompanying script independently checks (2), boundary moduli, the strip formula, the inverse, interior containment, and the removable derivative at $w = 1$ for $\omega = 1.2, 2, 5$. Boundary and inverse errors were below 3×10^{-15} ; the finite-difference derivative errors at step 10^{-6} were below 8×10^{-6} .

A targeted search through 13 August 2026 found the 2012/2014 standard Ritt square-function theory and the 2024 Ritt_E contour decomposition, but no source giving the exact generalized-Chebyshev map (1) or the coefficient-space obstruction above. This is evidence about search coverage, not a claim of bibliographic priority.

References

- [1] S. Fackler and M. Hauer, *Remarks on Ritt operators, their H^∞ functional calculus and associated square function estimates*, arXiv:2303.03022 (2023).
- [2] C. Le Merdy, *H^∞ functional calculus and square function estimates for Ritt operators*, Rev. Mat. Iberoam. 30 (2014), 1149–1190; arXiv:1202.0768.
- [3] O. Bouabdillah, *Square functions associated with Ritt_E operators*, arXiv:2410.22006 (2024).

Open question: For a given Ritt operator, the link between its H^∞ functional calculus and its associated square function estimates resembles the analogous link for sectorial (or strip type) operators A . In fact, the proofs we present are built up using the same abstract principles.

For both, Ritt and sectorial operators on Banach spaces of finite cotype, the functional calculus (if it is bounded) can be extended to so-called “bounded square functional” (or “quadratic”) H^∞ -calculus, verifying

$$(6.4) \quad \mathbb{E} \left\| \sum_n \gamma_n f_n(T) x \right\| \lesssim \sup_z \left(\sum_n |f_n(z)|^2 \right)^{1/2} \|x\|.$$

In the case of sectorial (or strip type) operators, the square functional calculus can be deduced directly from Theorem A.1 via integral representations that make direct use of “standard” square functions. In the case of Ritt operators however, this does not seem to work. The integral representation analogous to the sectorial situation would be

$$(6.5) \quad f(z) = \sum_k a_k \varphi_{m_1}(k, z) \varphi_{m_2}(k, z)$$

where the coefficients a_k do not depend on the variable z (unlike the formula (6.1) above). Despite this lack, the validity of (6.4) was shown in Le Merdy’s paper [19]. Instead of using the assumed (dual) square functions of the functions $\varphi_m(z)$, he rather bootstraps the square functional calculus from the (scalar) H^∞ -calculus by means of the Francks-McIntosh decomposition that produce a kind of “brute force square functions” for a collection of functions which are not explicitly known. Our integral representation theorem A.1 can then be applied to the Francks-McIntosh decomposition.

From this observation two question emerge: first, we would like to identify the (non-separable) subspaces

$$\mathcal{S}_{m_1, m_2} := \left\{ \sum_k a_k \varphi_{m_1}(k, z) \varphi_{m_2}(k, z) : (a_k) \in \ell_\infty \right\}$$

and their sum in $H^\infty(\text{Stolz}_w)$. Second, we expect that other, explicit square functions for Ritt operators are to be found that will, allow amongst others, general representation formulas. This question might be linked to the conformal map of Stolz regions to the unit disc, that seem not to be explicitly known, either.

APPENDIX A. TOOLS FROM ABSTRACT FUNCTIONAL CALCULUS THEORY

We cite in this section to abstract results borrowed from Haak-Haase [9]. We

Figure 1: The exact open-question subsection on PDF page 15 of arXiv:2303.03022.